PhD Research Proposal: Epistemic Awareness for AI Safety

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1 Introduction

Since the release of ChatGPT in 2023, AI systems have become increasingly integrated into our daily lives and workflows. They have evolved from personal writing assistants to research assistants. Today, they are also deployed as agents in business processes and autonomous decision-making pipelines. Currently, more than 78% of businesses worldwide use AI, a number expected to grow significantly in the coming years [1].

Importantly, these systems are moving toward higher levels of autonomy requiring less human supervision [2]. This shift promises significant benefits, bringing the long-awaited hope of material and cognitive abundance closer to reality. At the same time, it introduces critical challenges: ensuring reliability, safe reasoning, and trustworthy behavior. Real-world incidents, such as autonomous vehicle failures, confidently incorrect LLM output, and unsafe behavior in high-stakes settings, highlight the risks that remain with current AI systems [3].

A necessary component for building safe and trustworthy AI is the awareness of uncertainty [4, 5]. Current models often produce overconfident predictions even when they are incorrect, and struggle to recognize when they lack sufficient knowledge or evidence. This makes them vulnerable to distribution shifts or out-of-distribution (OOD) inputs and can also lead to models behaving unpredictably in new scenarios.

Addressing these issues requires a reexamination of the theoretical foundations of modern AI, which predominantly rely on reasoning with precise probabilistic formalisms. Such approaches, while powerful, impose strong assumptions that fundamentally enforce our machines to be "confident", such as the closed-world view in which the probability space is assumed to encode all relevant possibilities, and also lack the expressive flexibility to distinguish between indifference (equal degrees of belief) and ignorance (absence of justified belief) in probability assignments [6] (see Figure 1). These examples call for a more flexible, yet principled mathematical framework to represent and reason with uncertainty, such as imprecise probabilities [7].

Envisioning working with Professor Siu Lun Chau, my aim is to contribute to the emerging field of Imprecise Probabilistic Machine Learning (IPML), which seeks to endow AI systems with principled *"I don't know"* capabilities grounded in rigorous mathematical foundations. I believe that this is timely research to build a safe, responsible, and principled deployment of AI systems in society.

In Section 2, I present my research background and motivation, which lead into Section 3 on uncertainty in machine learning and my research vision. Section 4 then outlines the proposed methodology for my PhD research.

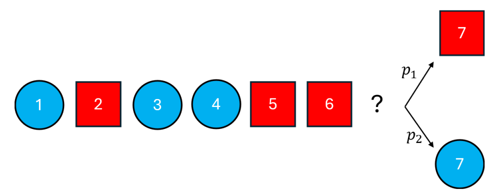

Figure 1: Consider the setup shown, where a model is tasked with predicting the next item in a sequence. Using standard probability formalisms, we cannot distinguish between two scenarios: (1) the model is confident that the sequence is genuinely random, assigning equal likelihood to each possible outcome ($p_1 = p_2 = 0.5$), or (2) the model simply lacks information and defaults to a 0.5 probability for each option. This ambiguity arises because probabilities are constrained to sum to 1, making it impossible to separate true randomness from epistemic uncertainty in this formalism.

2 Research Background and Motivation

Throughout my university years, I have actively explored five AI researches through various internships and research assistant roles. These experiences provided me with practical exposure to diverse problems and have deepened my appreciation of the challenges involved in the safe deployment of AI.

During my internship at A*STAR, I worked on medical recommendation systems with Dr. Pavitra Krishnaswamy, where incorrect predictions could critically affect patient care. At Continental Lab with Associate Professor Tianwei Zhang, I developed motion planning algorithms for robots to navigate in environments where small errors could lead to costly failures. At Temasek Laboratories, I analyzed IC hardware images for anomalies, where even minor mispredictions could compromise critical security assessments.

Moreover, while assisting S-Lab on large multimodal evaluations (lmms-evals) with Assistant Professor Ziwei Liu, an evaluation pipeline for multimodal models designed to integrate existing benchmarks and provide a modular interface for evaluating newly released models, I observed a notable absence of metrics for the model's uncertainty awareness, robustness, and out-of-distribution performance, highlighting how underexplored this area remains. This collaboration resulted in my first academic publication at NAACL 2025 [8], as well as a journal paper in IEEE TPAMI [9].

I am also currently preparing a publication with Huawei on personalized video understanding and reliable long-term memory, further strengthening my interest in reliable and uncertainty-aware AI systems.

Across these experiences, I observed a common pattern: challenges in safely deploying AI in high-stakes, real-world domains often arise from models being overconfident under distributional shift and limited knowledge. Only later did I come to recognize these failures as manifestations of what is commonly referred to as epistemic uncertainty, a notion that I formalize in Section 3. This realization fascinates me and motivates my PhD research vision: to develop principled frameworks for epistemic uncertainty awareness, enabling AI systems to reason safely when operating beyond the conditions encountered during training.

Driven by this realization, I sought theoretical and methodological frameworks to address uncertainty in a principled manner. Through a summer research engineering course on mechanistic interpretability [10], I gained a technical foundation to investigate internal representations and identify the neural circuits that drive model behavior.

These perspectives were further refined through discussions with Prof. Siu Lun Chau, whose work on epistemic uncertainty in machine learning closely aligns with the challenges I encountered in practice. In our discussions, Prof. Chau and I explored how mechanistic interpretability and game-theoretic frameworks could be systematically integrated with uncertainty modeling. This can potentially uncover how uncertainty is internally represented and propagated within learning systems, and how such representations could be leveraged to improve both explainability and robustness of AI models. This convergence of uncertainty quantification, interpretability, and multi-agent reasoning has helped crystallize a research direction that is both intellectually compelling and aligned with Prof. Chau's expertise [11], forming a strong foundation for a productive PhD collaboration.

3 Problem Statement and Research Vision

Building on this motivation, it is crucial to understand the specific challenges that limit current AI systems from reasoning safely under uncertainty.

Despite their remarkable capabilities, existing models often fail to recognize the limits of their own knowledge, leading to overconfident predictions and unpredictable behavior in high-stakes environments [12]. A central reason for this failure lies in how uncertainty itself is represented and interpreted within modern machine learning systems.

Aleatoric and Epistemic Uncertainty. Uncertainty in model predictions arises from two fundamentally different sources. **Aleatoric uncertainty** reflects irreducible noise inherent in the data-generating process, while **epistemic uncertainty** captures the model's ignorance due to insufficient, biased, or unrepresentative training data [13]. Distinguishing between these two forms of uncertainty is critical, as only epistemic uncertainty reflects limitations in the model's knowledge and can therefore serve as a meaningful signal for safety-aware decision-making.

3.1 Challenges in Current AI's Uncertainty Modeling

Eliciting Confidence. The theoretical distinction of the two uncertainty stated above is rarely respected by current methods [14]. While some approaches attempt to elicit confidence directly from a model's logits or via token-level sampling, these methods typically entangle aleatoric and epistemic uncertainty, preventing a clear attribution of reliability (see Figure 3). As a result, the confidence signals produced are difficult to interpret and unreliable for downstream decisions. Consequently, such estimates are shown to often be uncalibrated and inaccurate [15]. This entanglement represents a critical bottleneck for deploying AI systems in safety-critical settings, where knowing when not to act is as important as knowing what to predict.

Out-of-Distribution Behaviour. Poor uncertainty modeling becomes critically problematic under distribution shift. AI systems deployed in real-world settings inevitably encounter out-of-distribution (OOD) inputs that diverge from their training data. However, current models remain prone to overconfident misclassification, frequently attributing high probability to inputs that lie far outside the training manifold [16]. This phenomenon exposes a fundamental limitation of relying solely on observed performance: without explicit uncertainty awareness built in, we cannot distinguish genuine alignment from strategic compliance, underscoring its necessity in modern AI systems.

Model Interpretability. The challenges above are compounded by the opacity of modern AI systems. Despite rapid advances in deep learning and large language models, their internal reasoning and decision pathways largely remain black boxes. Existing explainability tools, such as saliency maps, attention rollout, and feature attribution, provide only partial and often brittle signals, particularly under distribution shifts or in complex real-world environments [17]. Unlike other safety-critical technologies, such as those in aviation, we still lack a principled understanding of what drives AI systems' behavior [18]. As AI systems exert increasing societal influence, this lack of interpretability undermines our ability to anticipate failures, assign responsibility, and establish trust. (see Figure 2)

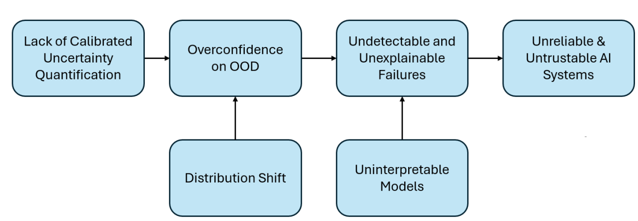

Figure 2: The absence of calibrated uncertainty quantification, when triggered by external distribution shifts, results in overconfident predictions on out-of-distribution (OOD) data. When these overconfident outputs are combined with uninterpretable model architectures, failures become both undetectable and unexplainable, ultimately undermining the reliability and trustworthiness of the deployed AI system.

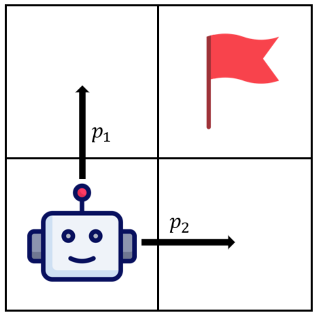

Figure 3: An AI agent choosing between two symmetric optimal actions, moving upward with probability p_1 and rightward with probability p_2, both derived from the model logits. While the assignment $p_1 = p_2 = 0.5$ is optimal, its high entropy is interpreted as low confidence under logit-based confidence elicitation. This illustrates a fundamental limitation of using logits for confidence estimation, even in language models: uncertainty arising from multiple equally valid options is conflated with epistemic uncertainty about the model's knowledge.

3.2 Research Vision

Taken together, these considerations motivate a research agenda focused on developing principled, uncertainty-aware, and interpretable AI systems that can reliably recognize the limits of their own knowledge and act safely under real-world uncertainty. I wish to explore uncertainty representations that explicitly preserve epistemic ambiguity under insufficient or biased evidence, rather than collapsing it into spurious confidence. By leveraging imprecise probabilistic models, the goal is to enable conservative reasoning, principled deferral, robustness under distributional shift, and interpretable attribution of uncertainty to epistemic versus aleatoric sources, thereby aligning decision-making with safety and interpretability requirements.

3.3 Implications and Broader Impact

The inability of AI systems to reliably model and act upon uncertainty is not merely a technical limitation, but a fundamental safety risk with real-world consequences.

Safety as a Prerequisite. AI systems that cannot identify the limits of their own knowledge are inherently unsafe, regardless of benchmark performance. Therefore, estimating when a model is in error is of great concern to AI Safety [3]. This research treats reliability as a core component of reasoning, enabling models to defer, abstain, or seek human oversight when epistemic uncertainty is high.

High-Stakes Deployment. In domains such as healthcare, finance, and defense, uncertainty-aware decision-making acts as a critical safety valve, reducing the risk of confident but erroneous judgments and protecting individuals from avoidable harm.

Governance and Alignment. As regulations increasingly require transparency and accountability, black-box confidence scores are insufficient. Principled uncertainty representations provide the auditability needed for regulatory compliance and support long-term alignment by promoting corrigibility, interpretability, and cautious behavior under uncertainty.

4 Methodology

4.1 Phase I: Developing a Scalable and Axiomatic Attribution Framework

Attribution Algorithm Development. I will first develop an efficient and scalable game-theoretic data attribution algorithm capable of identifying the specific training points responsible for model predictions and observed failure modes. Data attribution is paramount for trustworthy AI, offering direct insights into the fairness, safety, and provenance of the training data, often a more critical step than just feature attribution, as problematic data is the root cause of many model failures.

Axiomatic data valuation methods, such as the **Shapley Value** derived from Cooperative Game Theory [19], provide principled guarantees for contribution assessment. However, their direct application is severely restricted by combinatorial complexity. Given a value function $v : 2^D \to \mathbb{R}$ quantifying the contribution of a data subset, the Shapley value for a data point j is:

$$\phi_j = \sum_{S \subseteq D \setminus \{j\}} \mu(|S|)[v(S \cup \{j\}) - v(S)]$$

The necessity of iterating over all 2^n possible subsets S results in an exponential complexity of $O(2^n)$ with the number of data points (n).

This computational bottleneck is the central challenge to realizing truly principled trustworthy AI. Our core objective is to resolve this intractability by investigating novel mathematical approximation techniques aimed at reducing this complexity to at least $O(n^2)$ or better. Previous research has established and improved computational efficiency for computing Shapley values of **feature attribution** for product kernel methods [20], yet applying these rigorous, axiomatic methods to the data domain presents a renewed challenge. This breakthrough in computational tractability will ensure that axiomatic data attribution methods are scalable and practical for large-scale, real-world deployment.

Integration of Uncertainty. Crucially, I will extend this framework to rigorously incorporate *epistemic uncertainty* quantification, by adapting Prof Chau's work on Stochastic Shapley values [21] to data attribution. This involves analyzing how individual data instances contribute not merely to point predictions, but to the model's overall epistemic uncertainty. This extension allows the framework to directly inform the safety layer by attributing model ignorance to specific data scarcity or conflicts.

4.2 Phase II: Integrating Uncertainty Attribution for Learning-to-Defer Mechanisms

In Phase II, I shift the focus toward utilizing uncertainty attribution to inform the delegation of AI outputs. While uncertainty estimates provide a measure of model confidence, attribution allows us to understand *why* a model is uncertain, which is critical for determining if a human expert possesses the specific knowledge the model lacks.

From Rejection Learning to Deferral. The concept of a model abstaining from prediction was pioneered by Chow [22], who established the Bayes optimal "reject option" based on a fixed penalty for errors. While classical **rejection learning** [23] focuses on minimizing model error by withholding low-confidence predictions, the **Learning-to-Defer (L2D)** framework [24] generalizes this by considering the expertise of a specific collaborator. In L2D, the system does not simply reject a sample; it chooses to *defer* to a human expert whose predicted accuracy on that specific sample is factored into the loss. The objective minimizes:

$$\mathcal{L}_{\text{L2D}} = \sum_{i} \left[(1 - s_i)\ell(Y_i, \hat{Y}_{M,i}) + s_i \ell(Y_i, \hat{Y}_{D,i}) + s_i \gamma \right] \quad (1)$$

where s_i is the deferral decision, $\hat{Y}_{M,i}$ is the model prediction, and $\hat{Y}_{D,i}$ is the expert's prediction. By replacing Chow's constant rejection penalty with the human expert's loss $\ell(Y_i, \hat{Y}_{D,i})$, the framework optimizes for the joint performance of the human-AI team rather than the model in isolation.

While Chow [22] established that a posterior probability threshold is optimal for a fixed rejection penalty, this assumes a constant cost of abstention. In L2D, the "cost" is the expert's error, which varies across the input space. My approach moves beyond Chow's threshold by using uncertainty attribution to identify *why* the model is uncertain. By distinguishing between stochastic noise and epistemic gaps, the system can route tasks to specialized experts whose specific knowledge profiles complement the model's identified ignorance. This transforms the safety layer into an adaptive, multi-expert routing mechanism that optimizes for human-AI complementarity.

4.3 Phase III: Epistemic Fairness and AGI Safety

The long-term vision of this research is to establish a paradigm of **Epistemic Fairness**. I will define and formalize a new notion of fairness in AI by integrating the attribution tools of Phase I with the deferral mechanisms of Phase II. While traditional fairness focuses on the parity of outcomes, I propose to focus on *epistemic standing*, the model's right to render a decision. This builds on the concept of **hermeneutical injustice** [25], where a system causes harm by forcing a judgment in contexts it lacks the representative data to understand.

This approach also offers a scalable path for **AGI Alignment**. A primary risk in advanced AI is the "competence-alignment gap," where a system pursues goals efficiently but lacks the conceptual framework to recognize social or ethical nuances. By formalizing epistemic uncertainty as a refusal mechanism, this research helps address the **outer alignment problem** [26], which is the discrepancy between a human's true intentions and the mathematically specified objectives that an agent might over-optimize at the expense of unstated ethical or social constraints. Establishing these epistemic boundaries ensures that as systems become more capable, they remain tethered to human oversight in domains where their internal representations are sparse, fostering AI that is inherently corrigible and self-limiting [27].

5 Conclusion

This proposal addresses AI reliability by integrating uncertainty attribution with modern human-AI collaboration. By developing Stochastic Shapley values, I provide a mechanism to decompose model ignorance into actionable signals. These insights enable a nuanced Learning-to-Defer (L2D) framework that routes complex cases to specialized human experts, effectively balancing system autonomy with expert oversight. By grounding these technical advancements in the pursuit of hermeneutical justice, this work establishes a **foundation for Safe AGI**: ensuring the next generation of AI systems are not only efficient, but epistemically aware, fundamentally fair, and aligned with human values. Given the rapid integration of AI into high-stakes societal infrastructure, I believe this research is both timely and critical.

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